

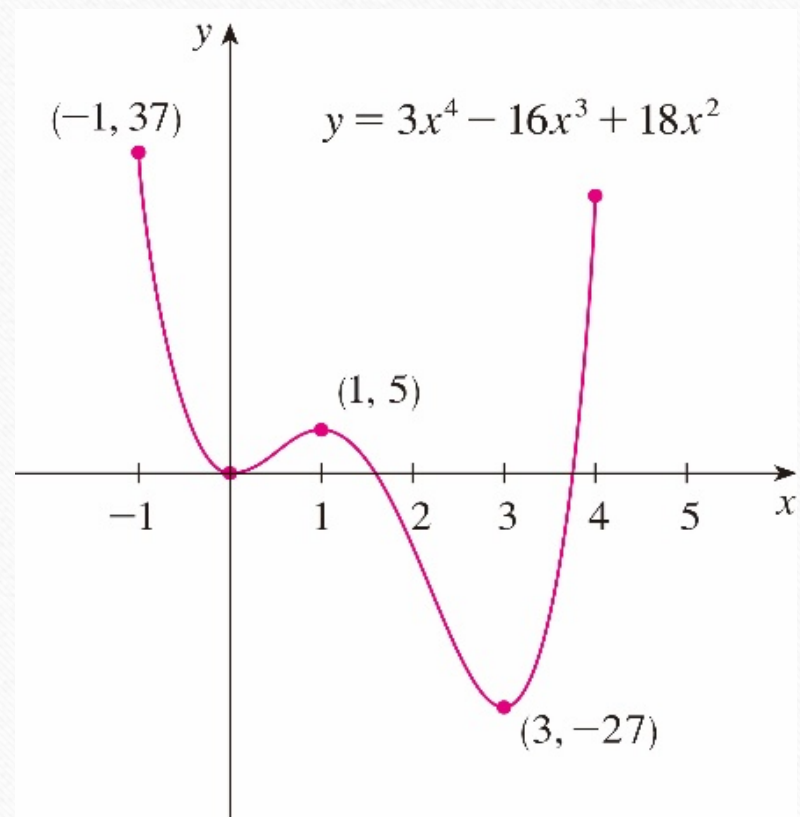
4-1 Extreme Values of Functions on Closed Intervals

師大工教一

Definition: Let f be a function with domain D . Then f has an **absolute maximum** value on D at a point c if $f(x) \leq f(c)$ for all x in D and an **absolute minimum** value on D at a point c if $f(x) \geq f(c)$ for all x in D .

global 整体的

Note: Maximum and minimum values are called **extreme values** of the function f . Absolute maxima or minima are also referred to as **global** maxima or minima.



Theorem 1—The Extreme Value Theorem

極值定理

If f is continuous on a closed interval $[a, b]$, then f attains both an absolute maximum value M and an absolute minimum value m in $[a, b]$.

That is, there are numbers x_1 and x_2 in $[a, b]$ with $f(x_1) = M, f(x_2) = m$, and $m \leq f(x) \leq M$ for every other x in $[a, b]$.

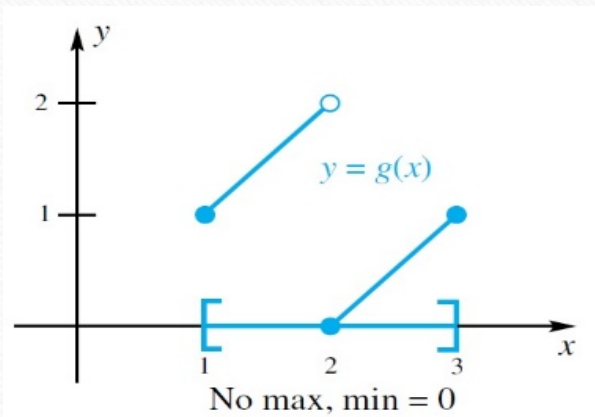
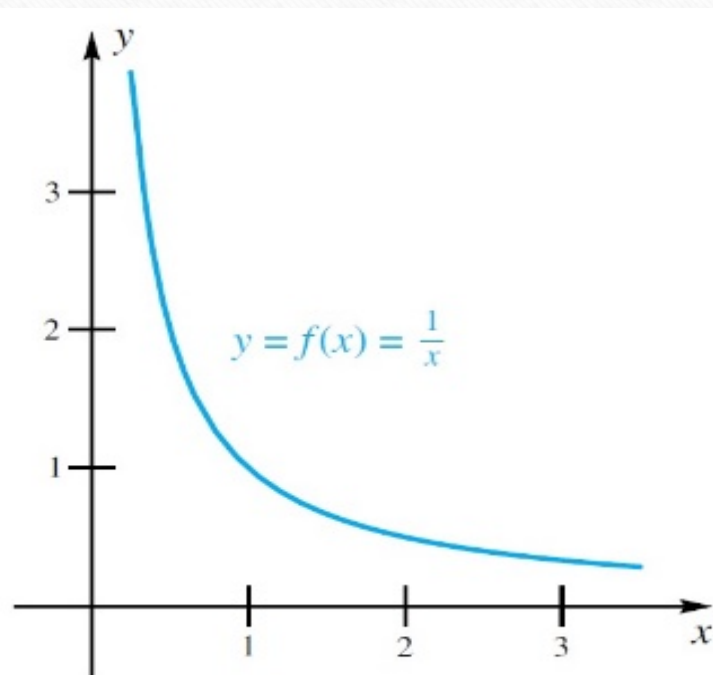


Figure 3



On $(0, \infty)$, no max or min

On $[1, 3]$, max = 1, min = $\frac{1}{3}$

On $(1, 3]$, no max, min = $\frac{1}{3}$

Figure 2

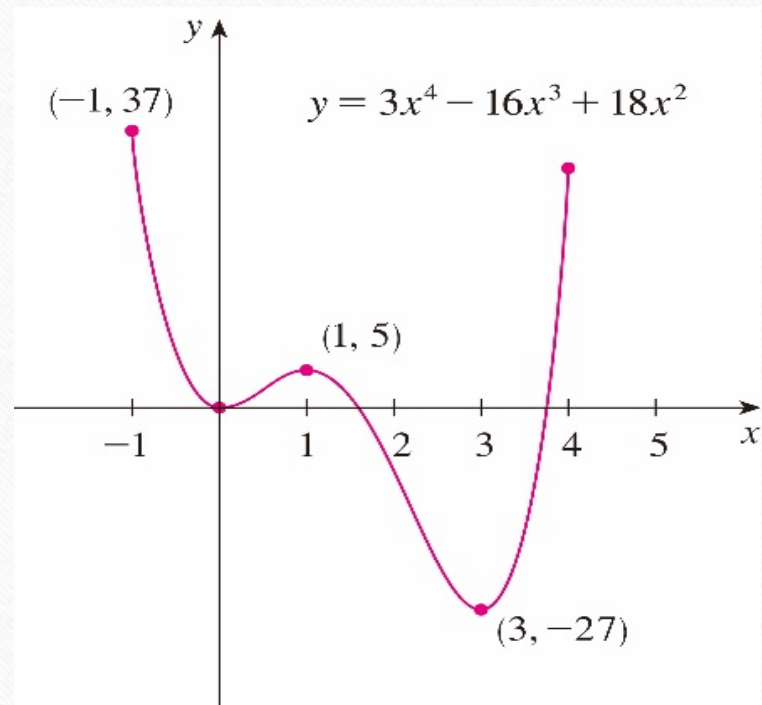


Figure 4.3(p232) shows that extreme values could happen at endpoints or interior points of the function.

局部極值

Local (Relative) Extreme Values

Definition: A function f has a **local maximum** value at a point c within its domain D if $f(x) \leq f(c)$ for all $x \in D$ lying in some open interval containing c .

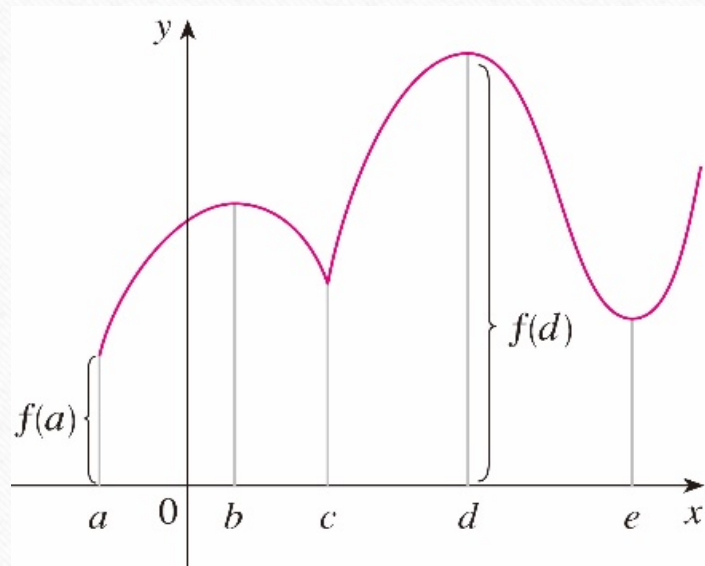
A function f has a **local minimum** value at a point c within its domain D if $f(x) \geq f(c)$ for all $x \in D$ lying in some open interval containing c .

Local extrema are also called **relative extrema**.

Finding Extrema

Theorem 2—The First Derivative Theorem for Local Extreme Values

If f has a local maximum or minimum value at an interior point c of its domain, and if f' is defined at c , then $f'(c) = 0$.



Definition: An interior point of the domain of a function f where f' is zero or undefined is a **critical point** of f .

臨界点

Finding the Absolute Extrema of a Continuous Function f on a Finite Closed Interval

1. Find all critical points of f on the interval.
2. Evaluate f at all critical points and endpoints.
3. Take the largest and smallest of these values.

Ex3(p234): Find the absolute maximum and minimum values of

$f(x) = 10x(2 - \ln x)$ on the interval $[1, e^2]$.

$$\begin{aligned} f'(x) &= 10(2 - \ln x) + 10x \left(-\frac{1}{x}\right) = 20 - 10 \ln x - 10 \\ &= 10 - 10 \ln x \stackrel{!}{=} 0 \end{aligned}$$

$$\begin{aligned} \ln x &= 1 \\ x &= e \leftarrow (\text{critical point}) \end{aligned}$$

$f(e) = 10e$ is the absolute maximum value.

$$f(1) = 20$$

$f(e^2) = 0$ is the absolute minimum value

Ex4(p235): Find the absolute maximum and minimum values of $f(x) = x^{\frac{2}{3}}$ on the interval $[-2, 3]$.

$$f'(x) = \frac{2}{3} x^{-\frac{1}{3}} \stackrel{\text{let}}{=} 0 \quad x=0 \text{ is a critical point}$$

$$x^{-\frac{1}{3}} = 0$$

$$x = 0$$

$f(0) = 0$ is absolute minimum value

$$f(-2) = \sqrt[3]{4}$$

$f(3) = \sqrt[3]{9}$ is absolute maximum value.

HW4-1

- **HW: 24,29,38,47,50,69,72**

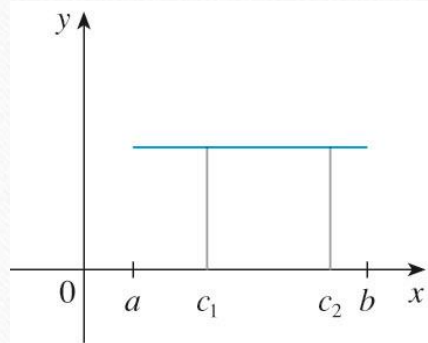
4-2 The Mean Value Theorem

均值定理

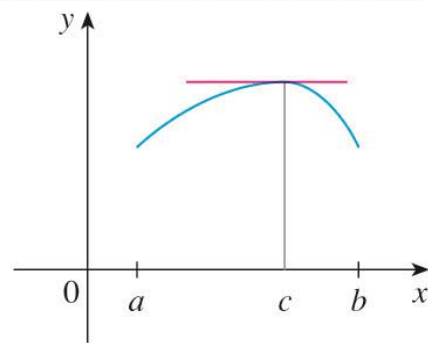
師大工教一

Theorem 3—Rolle's Theorem

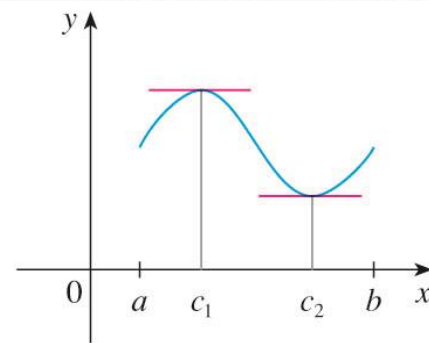
Suppose that $y = f(x)$ is continuous over the closed interval $[a, b]$ and differentiable at every point of its interval (a, b) . If $f(a) = f(b)$, then there is at least one number $c \in (a, b)$ such that $f'(c) = 0$.



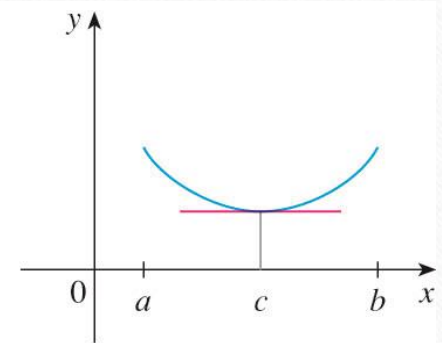
(a)



(b)



(c)



(d)

Ex1(p239) Show that the equation $x^3 + 3x + 1 = 0$ has exactly one real solution.

Rolle's Theorem

(H1) f is continuous on $[a, b]$

(H2) f is differentiable on (a, b)

(H3) $f(a) = f(b)$

(C) $\exists c \in (a, b)$ such that $f'(c) = 0$

Let $f(x) = x^3 + 3x + 1$

(i) $f(-1) = -3 < 0 < f(0) = 1$

$\therefore f$ is a polynomial

$\therefore f$ is continuous on $[-1, 0]$

By I.V.T, $\exists c \in (-1, 0)$ such that $f(c) = 0$, i.e., $c^3 + 3c + 1 = 0$.

$\therefore c$ is a real solution of $x^3 + 3x + 1$

(ii) Assume that $x_1 < x_2$ & x_1, x_2 are real solutions of $x^3 + 3x + 1 = 0$

Note that $f(x_1) = f(x_2) = 0$

Again, since f is a polynomial, f is continuous on $[x_1, x_2]$

& f is differentiable on (x_1, x_2)

By Rolle's Theorem, $\exists d \in (x_1, x_2)$

such that $f'(d) = 0$ $\rightarrow \leftarrow$

However, $f'(x) = 3x^2 + 3 \geq 3$ (it does not exist)

Therefore $f(x) = 0$ has exactly one real solution.

Theorem 4—The Mean Value Theorem

Suppose $y = f(x)$ is continuous over the closed interval $[a, b]$ and differentiable on the interval's interior (a, b) . Then there is at least one point

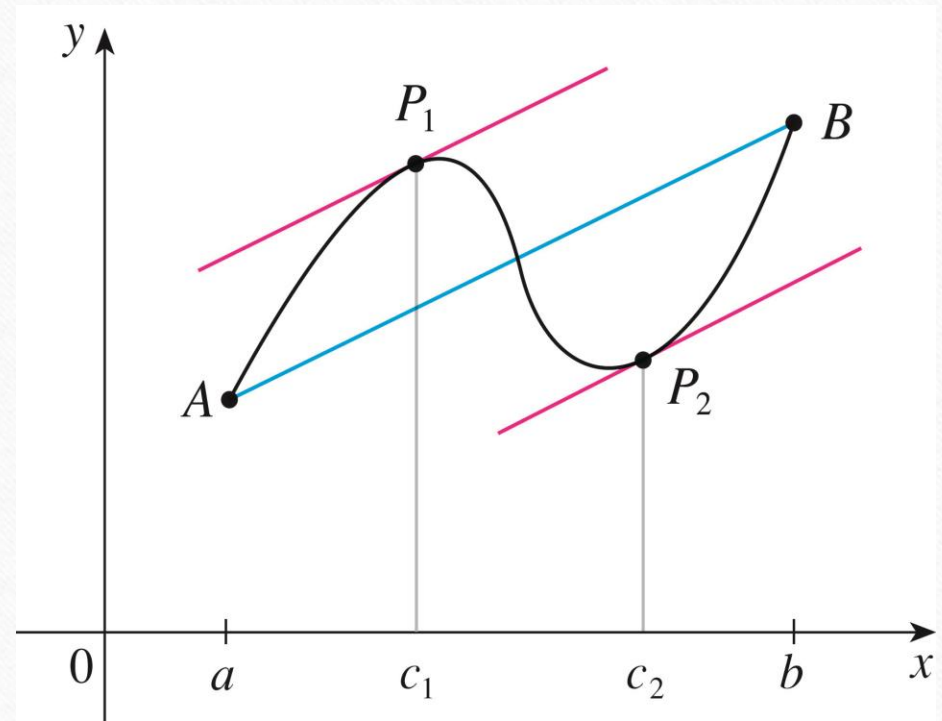
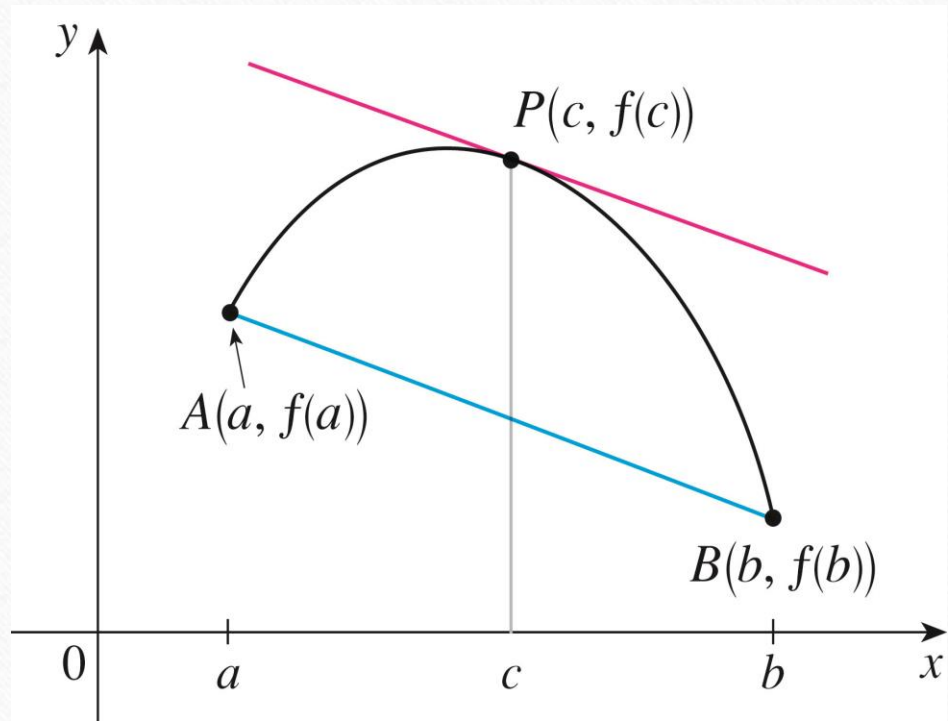
$$c \in (a, b) \text{ at which } \frac{f(b) - f(a)}{b - a} = f'(c).$$

(H₁) f is continuous on $[a, b]$

(H₂) f is differentiable on (a, b)

(C) $\exists c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$



Ex2(p241) Consider $f(x) = x^2$ over $[0, 2]$.

(i) f is continuous on $[0, 2]$.

(ii) f is differentiable on $(0, 2)$.

By Mean-Value Theorem, $\exists c \in (0, 2)$ such that $f'(c) = \frac{f(2) - f(0)}{2 - 0} = \frac{4}{2} = 2$.

Actually, $c = 1$. Note: In most cases, it's not easy to find c .

Corollary 1: If $f'(x) = 0$ at each point x of an open interval (a, b) , then

$f(x) = C$ for all $x \in (a, b)$, where C is a constant.

p.f: Let x_1, x_2 be any points
satisfying $a < x_1 < x_2 < b$
 $\therefore f'(x) = 0$ for all $x \in (a, b)$
 $\therefore f$ is differentiable on (x_1, x_2)
 $\therefore$ Differentiable implies Continuity
 $\therefore f$ is continuous on $[x_1, x_2]$
 By Mean Value Theorem
 $\exists c \in (x_1, x_2)$ such that $f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$
 $\therefore f'(c) = 0 \therefore f(x_1) = f(x_2)$
 $\therefore f(x) = C$ for all $x \in (a, b)$

Corollary 2: If $f'(x) = g'(x)$ at each point x of an open interval (a, b) , then

there exists a constant C such that $f(x) = g(x) + C$ for all $x \in (a, b)$. That

is, $f - g$ is a constant function on (a, b) .

Let $h(x) = f(x) - g(x)$,
 $h'(x) = f'(x) - g'(x) = 0 \quad x \in (a, b)$
 By Corollary 1, $h(x) = C$.
 $f(x) - g(x) = C$
 $f(x) = g(x) + C$

Ex4(p242) Find the function $f(x)$ whose derivative is $\sin x$ and whose graph pass through the point $(0, 2)$.

$$f'(x) = \sin x$$

$$(-\cos x)' = \sin x$$

By corollary 2, $\exists C$ such that

$$f(x) = -\cos x + C$$

$$f(0) = -\cos 0 + C = 2 \quad (\because \text{passing through } (0, 2))$$

$$\Rightarrow C = 3$$

$$\therefore f(x) = -\cos x + 3$$

HW4-2

- HW: 4,21,23,33,35,39,41

(107) 3. (10 pts) Does the nonlinear equation (非線性方程式) $e^{-3x} - x = 0$ have a unique solution (唯一解) in the interval $[0, 1]$? Give your reasons.

I.V.T

Rolle's

$$\text{Let } f(x) = e^{-3x} - x$$

$$(i) \quad f(0) = 1 - 0 = 1$$

$$f(1) = e^{-3} - 1 = \frac{1}{e^3} - 1 < 0$$

$$f(1) = \frac{1}{e^3} - 1 < 0 < f(0) = 1$$

$\therefore f$ is a polynomial

$\therefore f$ is continuous on $[1, 0]$

By I.V.T $\exists c \in (0, 1)$ such that
 $f(c) = 0$, $e^{-3c} - c = 0$

$\therefore c$ is a real solution of $e^{-3x} - x$

(ii) Assume that $x_1 < x_2$ & x_1, x_2
are real solution of $e^{-3x} - x = 0$

Note that $f(x_1) = f(x_2) = 0$

Again, since f is a polynomial,

f is continuous on $[x_1, x_2]$

& f is differentiable on (x_1, x_2)

By Rolle's Theorem $\exists d \in (x_1, x_2)$

such that $f'(d) = 0$

However, $f'(x) = -3e^{-3x} - 1 < -1$, $\nexists d$ (it does not exist)

Therefore, $f(x) = 0$ has a exactly solution

單調的

4-3 Monotonic Functions and the First Derivative Test

一階導數判別法

師大工教一

Increasing Functions and Decreasing Functions

Corollary 3 Suppose that f is continuous on $[a, b]$ and differentiable on

$$f'(x) > 0, \text{ on } (a, b)$$

$\Rightarrow f$ is increasing (递增, $\nearrow$) on $[a, b]$

(a, b) .

$$f'(x) < 0, \text{ on } (a, b)$$

$\Rightarrow f$ is decreasing (递减, $\searrow$) on $[a, b]$

If $f'(x) > 0$ at each point $x \in (a, b)$, then f is increasing on $[a, b]$.

If $f'(x) < 0$ at each point $x \in (a, b)$, then f is decreasing on $[a, b]$.

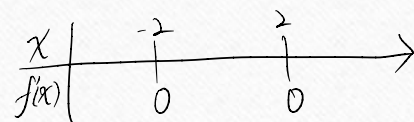
Note: A function that is either increasing on an interval or decreasing on an interval is said to be **monotonic** on the interval.

Ex1(p247) Find the critical points of $f(x) = x^3 - 12x - 5$ and identify the open intervals on which f is increasing and those on which f is decreasing.

$$f'(x) = 3x^2 - 12 \stackrel{\text{let}}{=} 0$$

$$x^2 = 4$$

$$x = \pm 2, \text{ (critical points)}$$



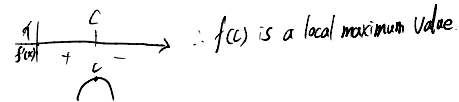
increasing ($\nearrow$) : $(-\infty, -2), (2, \infty)$

decreasing ($\searrow$) : $(-2, 2)$

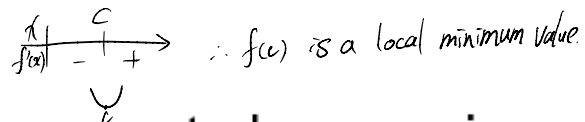
First Derivative Test for Local Extrema

Suppose that c is a critical point of a continuous function f , and that f is differentiable at every point in some interval containing c except possibly at c itself. Moving across this interval from left to right.

1. if f' change from positive to negative at c , then f has a local maximum at c .

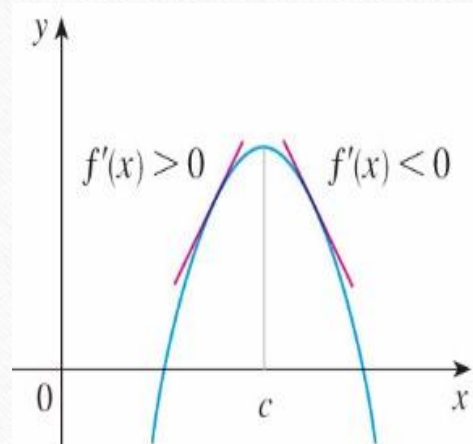


2. if f' change from negative to positive at c , then f has a local minimum at c .

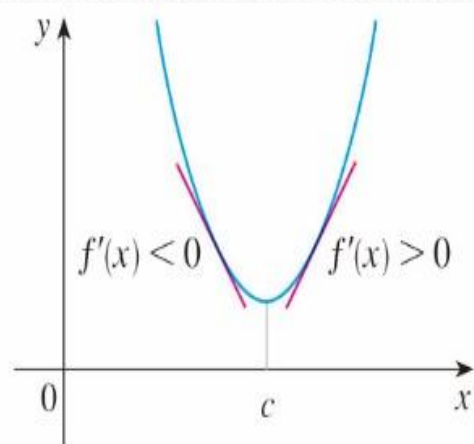


3. if f' does not change sign at c (that is, f' is positive on both sides of c or negative on both sides), then f has no local extremum at c .

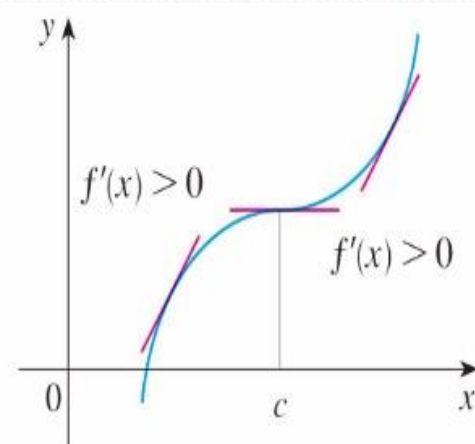




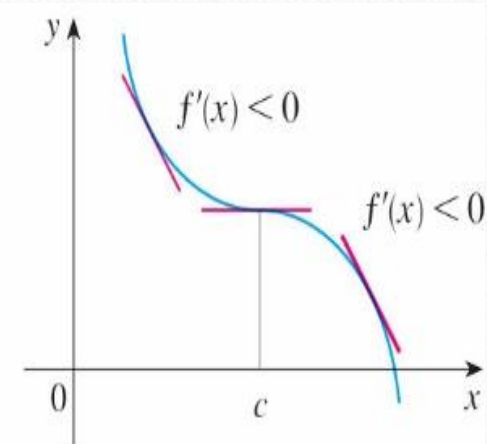
(a) Local maximum



(b) Local minimum



(c) No maximum or minimum



(d) No maximum or minimum

Ex3 Find the critical points of $f(x) = (x^2 - 3)e^x$. Identify the open intervals on which f is increasing and those on which it is decreasing. Find the function's local and absolute extreme values.

$$\begin{aligned} f'(x) &= 2xe^x + (x^2 - 3)e^x \\ &= e^x(x^2 + 2x - 3) \\ &= e^x(x + 3)(x - 1) \stackrel{!}{=} 0 \end{aligned}$$

$$x = -3, 1 \text{ (critical point)}$$

$$\begin{array}{c|ccc} x & -3 & & 1 \\ \hline f'(x) & + & 0 & - & 0 & + \end{array}$$

$$\nearrow : (-\infty, -3), (1, \infty)$$

$$\searrow : (-3, 1)$$

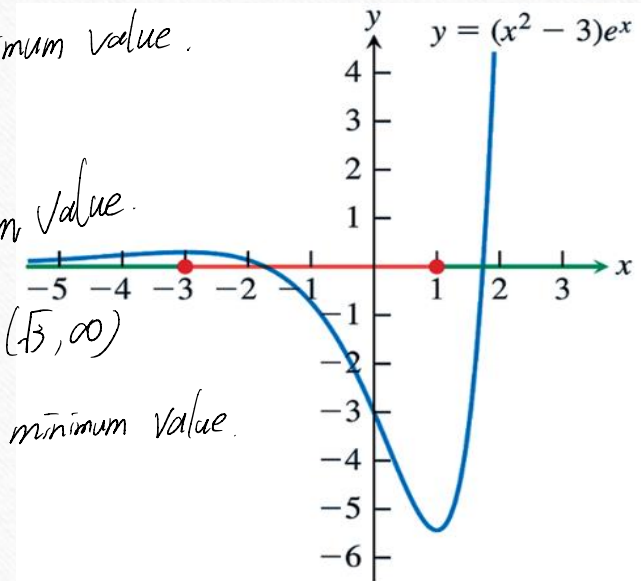
$f(-3) = 6 \cdot e^{-3}$ is a local maximum value.
 $f(1) = -2e$ is a local minimum value.

$$\lim_{x \rightarrow \infty} (x^2 - 3)e^x = \infty$$

No Absolute Maximum value.

$$\therefore f(x) > 0 \text{ when } x \in (-\infty, -3), (1, \infty)$$

$$\therefore f(1) = -2e \text{ is the Absolute minimum value.}$$

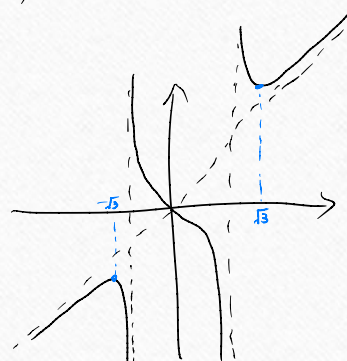
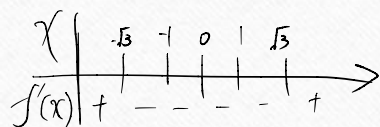


Ex(107 考古題#4(a)) For the real-valued function(實值函數) $f(x) = \frac{x^3}{x^2 - 1}, \dots$

(a) Determine the open intervals on which f is increasing(遞增) or decreasing(遞減).

$$f'(x) = \frac{3x^2(x^2-1) - x^3(2x)}{(x^2-1)^2} = \frac{3x^4 - 3x^2 - 2x^4}{(x^2-1)^2} = \frac{x^4 - 3x^2}{(x^2-1)^2}$$

$$= \frac{x^2(x^2-3)}{(x^2-1)^2} = \frac{x^2(x+\sqrt{3})(x-\sqrt{3})}{(x^2-1)^2}$$



$$\nearrow : (-\infty, -\sqrt{3}), (\sqrt{3}, \infty)$$

$$\searrow : (-\sqrt{3}, -1), (-1, 0), (0, 1), (1, \sqrt{3})$$

HW4-3

-
- HW: 21,26,43,53,77,80

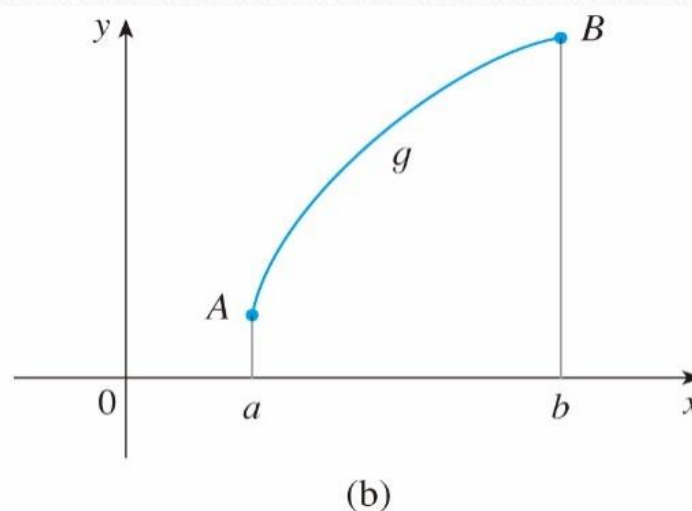
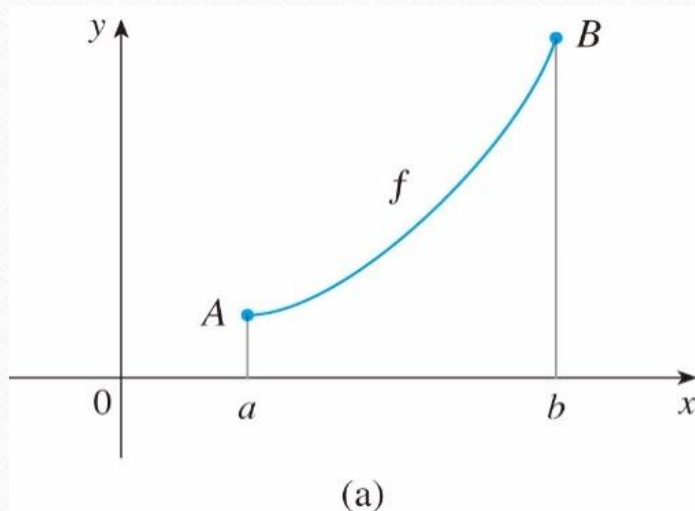
4-4 ^{凹性} Concavity and Curve Sketching

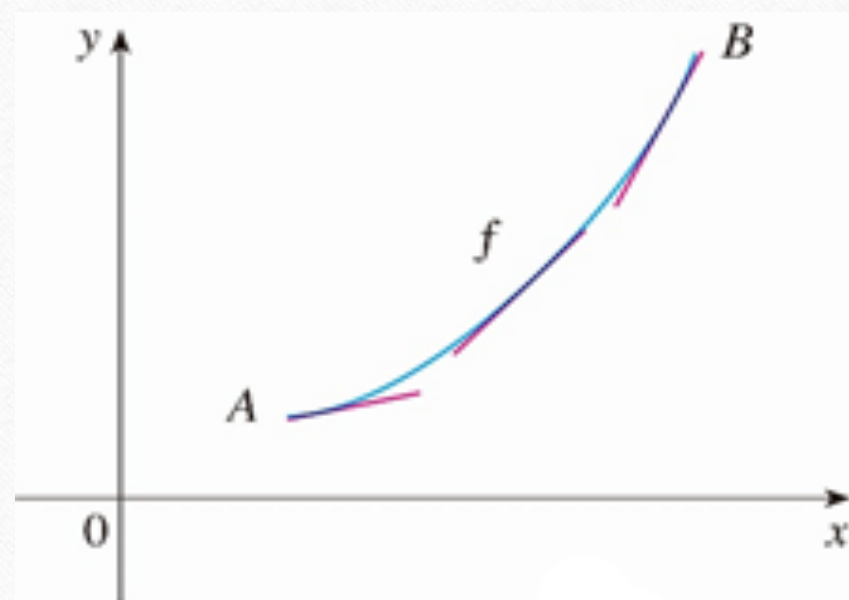
師大工教一

Concavity

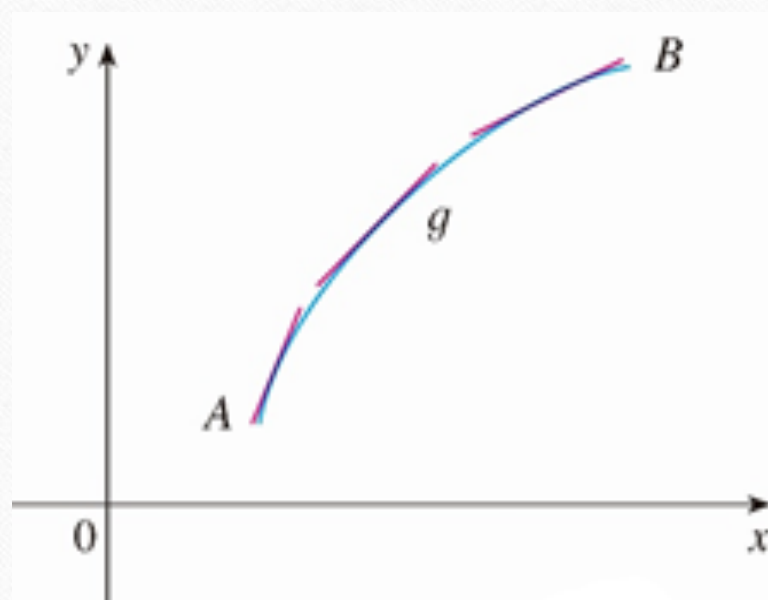
Definition The graph of a differentiable function $y = f(x)$ is

- (a) **concave up** on an interval I if f' is increasing on I .
(b) **concave down** on an interval I if f' is decreasing on I .





(a) Concave up



(b) Concave down

The Second Derivative Test for Concavity

Let $y = f(x)$ be twice differentiable on an interval I .

1. If $f'' > 0$ on I , the graph of f over I is concave up.
2. If $f'' < 0$ on I , the graph of f over I is concave down.

Ex1(p252) Discuss the concavity of the functions (a) $y = x^3$ (b) $y = x^2$

(a) $y' = 3x^2$

$y'' = 6x \stackrel{?}{=} 0$

$x = 0$

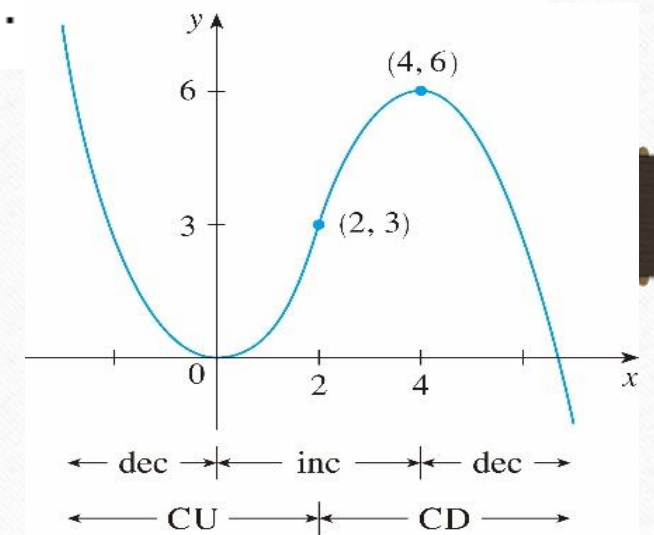
x | 0
| $-$ 0 $+$
| $f''(x)$
concave up: $(0, \infty)$
concave down: $(-\infty, 0)$

(b) $y' = 2x$
 $y'' = 2$

concave up: $(-\infty, \infty)$
concave down: None

Point of Inflection 反曲點 Inflection point

Definition A point $(c, f(c))$ where the graph of a function has a tangent line and where the concavity changes is a **point of inflection**.



At a point of inflection $(c, f(c))$, either $f''(c) = 0$ or $f''(c)$ fails to exist.

Ex3 Determine the concavity and find the inflection points of the function

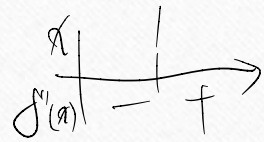
$$f(x) = x^3 - 3x^2 + 2.$$

$$f'(x) = 3x^2 - 6x$$

$$f''(x) = 6x - 6 \stackrel{\text{let}}{=} 0$$

$$x = 1$$

$(1, 0)$ is an inflection point.



concave up: $(1, \infty)$

concave down: $(-\infty, 1)$

Theorem 5—**Second Derivative Test for Local Extrema**

Suppose f'' is continuous on an open interval that contains $x = c$.

1. If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at $x = c$.
2. If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at $x = c$.
3. If $f'(c) = 0$ and $f''(c) = 0$, then the test fails. The function f may have a local maximum, a local minimum, or neither.

HW4-4

- HW: 11,17,35,50,52,58,61

(110) 8. Given a function $f(x) = x^4 - 2x^2$ on the interval $[-2, 3]$.

- (a) Find open intervals where f is increasing or decreasing. (6 pts)
- (b) Find the relative extrema of f . (10 pts)
- (c) Determine open intervals where f is concave upward or downward. (6 pts)

$$(a) f'(x) = 4x^3 - 4x = 4x(x^2 - 1) = 4x(x+1)(x-1) \stackrel{\text{let}}{=} 0$$

$$x = 0, -1, 1 \text{ (critical point)}$$

x	-1	0	1
$f'(x)$	-	+	-

 $\nearrow: (1, \infty), (-1, 0)$
 $\searrow: (-\infty, -1), (0, 1)$

(b) $f(-1) = f(1) = -1$ is a local minimum value.
 $f(0) = 0$ is a local maximum value.

$$(c) f''(x) = 12x^2 - 4 \stackrel{\text{let}}{=} 0$$

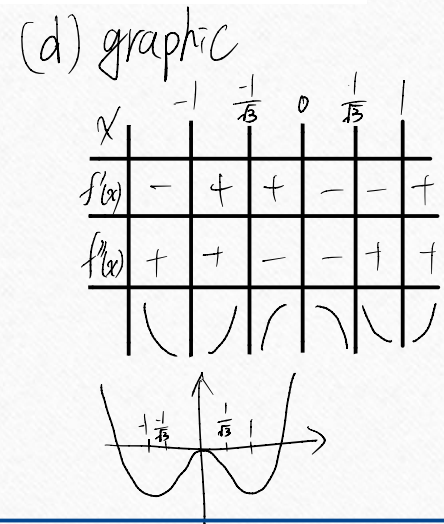
$$4(3x^2 - 1) = 0$$

$$4(\sqrt{3}x + 1)(\sqrt{3}x - 1) = 0$$

$$x = \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}$$

x	$-\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$
$f''(x)$	+	-

concave up: $(-\infty, -\frac{1}{\sqrt{3}}), (\frac{1}{\sqrt{3}}, \infty)$
 concave down: $(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$



Ex(108 年考古題#5) Consider the function $f(x) = x^{\frac{2}{3}}(x^2 - 1) = x^{\frac{8}{3}} - x^{\frac{2}{3}}$

(a) Find the open intervals on which f is increasing(遞增) or decreasing(遞減).

(b) Find the open intervals on which the graph of f is concave upward(凹口向上) or concave downward(凹口向下).

(c) Locate the point of inflection(反曲點) if it exists.

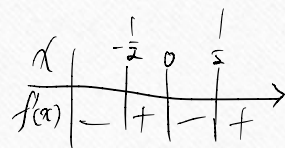
$$(a) f'(x) = \frac{8}{3}x^{\frac{5}{3}} - \frac{2}{3}x^{-\frac{1}{3}} \stackrel{!}{=} 0$$

$$\frac{8}{3}x^{\frac{5}{3}} - \frac{2}{3}x^{-\frac{1}{3}} = 0$$

$$\frac{8\sqrt[3]{x^5}}{3} - \frac{2}{3\sqrt[3]{x}} = 0$$

$$\frac{8\sqrt[3]{x^5} - 2}{3\sqrt[3]{x}} = \frac{8x^{\frac{5}{3}} - 2}{3\sqrt[3]{x}} = \frac{2(4x^{\frac{5}{3}} - 1)}{3\sqrt[3]{x}} = \frac{2(2x+1)(2x-1)}{3\sqrt[3]{x}} = 0$$

$$x = -\frac{1}{2}, \frac{1}{2}, 0$$



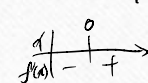
$$\nearrow: (-\frac{1}{2}, \infty), (-\frac{1}{2}, 0)$$

$$\searrow: (-\infty, -\frac{1}{2}), (0, \frac{1}{2})$$

$$(b) f''(x) = \frac{40}{9}x^{\frac{2}{3}} + \frac{2}{9}x^{-\frac{4}{3}}$$

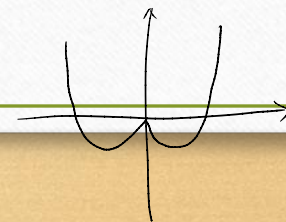
$$= \frac{40\sqrt[3]{x^2}}{9} + \frac{2}{9} \cdot \frac{1}{\sqrt[3]{x^4}}$$

$$= \frac{40\sqrt[3]{x^2} \cdot \sqrt[3]{x^4} + 2}{9\sqrt[3]{x^4}} = \frac{40x^2 + 2}{9\sqrt[3]{x^4}}$$



concave up: $(-\infty, 0), (0, \infty)$
concave down: None

(c) No inflection pt.



4-5 Indeterminate Forms and L'Hôpital's Rule

師大工教一

Indeterminate Form $\frac{0}{0}$

Theorem 6—L'Hôpital's Rule

Suppose that $f(a) = g(a) = 0$, that f, g are differentiable on an open interval I containing a , and that $g'(x) \neq 0$ on I if $x \neq a$. Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

assuming that the limit on the right side of the equation exists.

Ex1(p265) Find the following limit: (b) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$ (d) $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$

$$(b) \text{ origin } \left(\frac{0}{0}\right) = \lim_{x \rightarrow 0} \frac{\frac{1}{2\sqrt{1+x}}}{1} = \frac{1}{2}$$

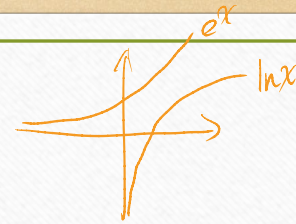
$$(d) \left(\frac{0}{0}\right) = \lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2} = \lim_{x \rightarrow 0} \frac{\sin x}{6x} = \lim_{x \rightarrow 0} \frac{\cos x}{6} = \frac{1}{6}$$

103 1 (c) (5 points) $\lim_{x \rightarrow 0} \frac{4(x - \sin(x))}{x^3} = ? = \lim_{x \rightarrow 0} \frac{4(1 - \cos x)}{3x^2} = \lim_{x \rightarrow 0} \frac{4(\sin x)}{6x} = \lim_{x \rightarrow 0} \frac{4\cos x}{6} = \frac{2}{3}$

Ex2(p266) Find $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x + x^2}$. (Be careful. Check the limit after every applications.)

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x + x^2} = \lim_{x \rightarrow 0} \frac{\sin x}{1 + 2x} = 0$$

Indeterminate Forms $\frac{\infty}{\infty}, \infty \cdot 0, \infty - \infty$



$$\lim_{x \rightarrow \infty} e^x = \infty$$

$$\lim_{x \rightarrow \infty} \ln x = \infty$$

$$\lim_{x \rightarrow \infty} e^x = \lim_{x \rightarrow \infty} e^{-x} = 0$$

$$\lim_{x \rightarrow 0^+} \ln x = -\infty$$

Ex4(p266) Find the limit (b) $\lim_{x \rightarrow \infty} \frac{\ln x}{2\sqrt{x}}$

$$\text{b) } \lim_{x \rightarrow \infty} \frac{\ln x}{2\sqrt{x}} \left(\frac{\infty}{\infty} \right) = \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{\sqrt{x}}} = \lim_{x \rightarrow \infty} \frac{\sqrt{x}}{x} = \lim_{x \rightarrow \infty} \frac{\frac{1}{\sqrt{x}}}{1} = 0$$

109 4 (d) $\lim_{t \rightarrow 0} \ln \sqrt{(1+t)^{\frac{1}{3t}}} = \lim_{t \rightarrow 0} \ln (1+t)^{\frac{1}{6t}} = \lim_{t \rightarrow 0} \frac{1}{6t} \ln(1+t) \left(\frac{0}{0} \right) = \lim_{t \rightarrow 0} \frac{\frac{1}{1+t}}{\frac{1}{6}} = \frac{1}{6}$

Ex5(p267) Find the limit (b) $\lim_{x \rightarrow 0^+} \sqrt{x} \ln x \ (0 \cdot (-\infty)) = \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{\sqrt{x}}} = \lim_{x \rightarrow 0^+} \frac{\ln x}{x^{\frac{1}{2}}} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{2}x^{\frac{3}{2}}} = \lim_{x \rightarrow 0^+} \frac{-2x^{\frac{3}{2}}}{x} = 0$

111 2

(c) $\lim_{x \rightarrow 0} \left(\frac{1}{x\sqrt{1+x}} - \frac{1}{x} \right) = \lim_{x \rightarrow 0} \frac{1 - \sqrt{1+x}}{x\sqrt{1+x}} \left(\frac{0}{0} \right) = \lim_{x \rightarrow 0} \frac{\frac{-1}{2\sqrt{1+x}}}{\sqrt{1+x} + \frac{x}{2\sqrt{1+x}}} = \lim_{x \rightarrow 0} \frac{-1}{2(1+x)+x} = -\frac{1}{2}$

✱ However 就是要變成 $\frac{0}{0}$ 或 $\frac{\infty}{\infty}$, 再用羅必達

Indeterminate Powers

If $\lim_{x \rightarrow a} \ln f(x) = L$, then $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} e^{\ln f(x)} = e^L$. Here a may be either finite or infinite.

Ex7(p268) Find the limit $\lim_{x \rightarrow 0^+} (1+x)^{\frac{1}{x}}$

$$= \lim_{x \rightarrow 0^+} \ln(1+x)^{\frac{1}{x}}$$

$$= \lim_{x \rightarrow 0^+} \frac{1}{x} \ln(1+x) \quad (\infty \cdot 0)$$

$$= \lim_{x \rightarrow 0^+} \frac{\frac{1}{1+x}}{x} \left(\frac{0}{0} \right) = \lim_{x \rightarrow 0^+} \frac{1}{1+x} = 1 \quad e' = e_{\neq}$$

105分 1 (d) $\lim_{x \rightarrow 0} (1+3x)^{\frac{1}{x}} \rightarrow \lim_{x \rightarrow 0} \frac{1}{x} \ln(1+3x) \quad (\infty \cdot 0) = \lim_{x \rightarrow 0} \frac{\frac{3}{1+3x}}{x} \left(\frac{0}{0} \right) = \lim_{x \rightarrow 0} \frac{3}{1+3x} = 1$
 $e' = e_{\neq}$

HW4-5

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- HW:5,16,21,30,35,37,53,58,59,71

4-8 Antiderivatives

師大工教一

Finding Antiderivatives

Definition: A function F is called an **antiderivative** of f on an interval I if $F'(x) = f(x)$ for all x in I .

Ex1(p289) Find an antiderivative of each of the following functions.

(a) $f(x) = 2x$ (b) $g(x) = \cos x$ (c) $h(x) = \frac{1}{x} + 2e^{2x}$

$$F(x) = x^2$$

$$G(x) = \sin x$$

$$H(x) = \ln|x| + e^{2x}$$

Theorem: If F is an antiderivative of f on an interval I , then the most general antiderivative of f on I is $F(x) + C$ where C is an arbitrary constant.

Theorem: If $F'(x) = G'(x)$, $\forall x \in I$,
then $F(x) = G(x) + C$, $x \in I$.

Ex2(p290) Find an antiderivative of $f(x) = 3x^2$ that satisfies $F(1) = -1$.

$$F(x) = x^3 - 2$$

Table 4.2 Antiderivative formulas

Recite the formulas, or at least know how to derive it!

Ex3(p208) Find the general antiderivative of each of the following functions,

(a) $f(x) = x^5$

$$F(x) = \frac{1}{6}x^6 + C$$

(b) $g(x) = \frac{1}{\sqrt{x}}$

$$G(x) = 2x^{\frac{1}{2}} + C$$

(c) $h(x) = \sin 2x$

$$H(x) = -\frac{1}{2}\cos 2x + C$$

(d) $i(x) = \cos \frac{x}{2}$

$$I(x) = 2\sin \frac{x}{2} + C$$

(e) $j(x) = e^{-3x}$

$$J(x) = -\frac{1}{3}e^{-3x} + C$$

(f) $k(x) = 2^x$

$$K(x) = \frac{2^x}{\ln 2} + C$$

Initial Value Problems and Differential Equations

Ex(#98, p297) Solve the initial value problem $\frac{ds}{dt} = \cos t + \sin t$, $s(\pi) = 1$.

$$S(t) = \sin t - \cos t + C$$

$$C = 0$$

$$S(t) = \sin t - \cos t$$

反導函數 = 不定積分

Indefinite Integrals

Definition The collection of all antiderivatives of f is called the **indefinite**

integral of f with respect to x ; it is denoted by $\int f(x)dx$. The symbol $\int$

is an **integral sign**. The function f is the **integrand** of the integrals, and x is the **variable of integration**.

Ex(#70,p296) $\int \frac{\csc \theta}{\csc \theta - \sin \theta} d\theta$

$$= \int \frac{\frac{1}{\sin \theta}}{\frac{1}{\sin \theta} - \sin \theta} d\theta = \int \sec^2 \theta d\theta = \tan \theta + C$$

$$= \int \frac{1}{1 - \sin^2 \theta} d\theta$$

$$= \int \frac{1}{\cos^2 \theta} d\theta$$

$$\int f(x)dx = F(x) + C$$

$\int$: integral sign 積分符號

f : integrand 被積分函數

HW4-8

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- HW: 26,29,39,46,52,67,76,78